

对偶理论：一般的约束优化问题

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad c_i(x) \leq 0 \quad i \in \mathcal{I} \quad ; \quad c_i(x) = 0, \quad i \in \mathcal{E}$$

$$\text{可行域 } \mathcal{X} = \{x \in \mathbb{R}^n \mid c_i(x) \leq 0 \quad i \in \mathcal{I} \quad ; \quad c_i(x) = 0, \quad i \in \mathcal{E}\}$$

Lagrange 函数：设 $|\mathcal{I}| = m$, $|\mathcal{E}| = p$

$$\mathcal{L}(x, \lambda, v) = f(x) + \sum_{i \in \mathcal{I}} \lambda_i c_i(x) + \sum_{i \in \mathcal{E}} v_i c_i(x)$$

弱对偶 $\leq f(x)$ 来控制 λ_i 正负 $\Rightarrow \lambda_i \geq 0 \quad \forall i \in \mathcal{I}$

考虑 $g: \mathbb{R}_+^m \times \mathbb{R}^p \rightarrow [-\infty, +\infty]$, $g(\lambda, v) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, v)$

Thm: 若 $\lambda \geq 0$, 则 $g(\lambda, v) \leq p^* \Rightarrow \underline{\sup_{\lambda \geq 0} g(\lambda, v)} \leq p^*$

称 q^* 为 p^* 的最优下界, $p^* - q^*$ 为对偶间隙 q^*

★ Lagrange 对偶问题为凸优化问题

$$\underbrace{\inf_x \sup_{\lambda \geq 0} \mathcal{L}(x, \lambda, v)}_{(P)} \geq \underbrace{\sup_{\lambda \geq 0} \inf_x \mathcal{L}(x, \lambda, v)}_{(D)}$$

$$\text{Exp: } \min x^T x \quad \text{s.t. } Ax=b$$

$$\mathcal{L}(x, v) = x^T x + v^T (Ax - b)$$

$$\text{对偶问题: } g(v) = \inf_x \mathcal{L}(x, v) = \mathcal{L}(-\frac{1}{2}A^T v, v) = -\frac{1}{4}v^T A A^T v - b^T v$$

$$\Rightarrow x^T x \geq p^* = -\frac{1}{4}v^T A A^T v - b^T v$$

LP 对偶:

$$\min_x c^T x \quad \text{s.t. } Ax=b, \quad x \geq 0$$

$$\mathcal{L}(x, s, v) = c^T x + v^T (Ax - b) - s^T x = -b^T v + (A^T v - s + c)^T x$$

$$g(s, v) = \inf_x \mathcal{L}(x, s, v) = \begin{cases} -b^T v & A^T v - s + c = 0 \\ -\infty & \text{else} \end{cases}$$

$$(D): \max_{s, v} -b^T v \quad \text{s.t. } A^T v - s + c = 0 \quad s \geq 0$$

若保留 $x \geq 0$:

线性对偶的对偶:

$$\max_y b^T y \quad \text{s.t.} \quad A^T y \leq c$$

$$\mathcal{L}(y, x) = -b^T y + x^T (A^T y - c) \quad (x \geq 0)$$

$$g(x) = \inf_y \mathcal{L}(y, x) = \begin{cases} -c^T x & Ax = b \\ -\infty & \text{else} \end{cases}$$

$$\Rightarrow \begin{aligned} & \max_x -c^T x \\ & \text{s.t.} \quad Ax = b \quad x \geq 0 \end{aligned}$$

故 LP 对偶的对偶 $\rightarrow$ 原问题 $\Rightarrow$ 若 LP 有最优解 $\Rightarrow$ LP(D) 有最优解, 且最优值相同.

Exp: $\min_x \|x\| \quad \text{s.t.} \quad Ax = b$

$$g(v) = \inf_x (\|x\| - v^T (Ax - b)) = \begin{cases} b^T v & \|A^T v\|_* \leq 1 \\ -\infty & \text{else} \end{cases}$$

其中 $\|v\|_* = \sup_{\|u\| \leq 1} u^T v$

(D): $\max_v b^T v \quad \text{s.t.} \quad \|A^T v\|_* \leq 1$

(WD): $p^* \geq b^T v \quad \text{若} \quad \|A^T v\|_* \leq 1$

一个问题不同形式的对偶可能差异很大.

· 引入新变量

· 显式约束 $\leftrightarrow$ 隐式约束

· 改变目标

$$\text{Exp: } \min_x f_0(Ax+b) \rightarrow \begin{cases} \min_y f_0(y) \\ y = Ax+b \end{cases} \xrightarrow{(D)} g(v) = \inf_{x,y} f_0(y) + v^T(Ax+b-y) = \begin{cases} \inf_y f_0(y) + v^T b - v^T y & A^T v = 0 \\ -\infty & \text{else} \end{cases} = -f_0^*(v) + b^T v$$

定义(共轭函数) : $f^*(y) = \sup_{x \in \text{dom} f} y^T x - f(x)$

$$\text{Exp: } \min_x \frac{1}{2} \|Ax-b\|^2 + \mu \|x\|_1 \Leftrightarrow \min_x \frac{1}{2} \|r\|^2 + \mu \|x\|_1 \quad \text{s.t. } r = Ax-b$$

$$\mathcal{L}(x, r, \lambda) = \frac{1}{2} \|r\|^2 + \mu \|x\|_1 + \lambda^T (r - Ax + b)$$

$$\Rightarrow g(\lambda) = \inf_{x,r} \mathcal{L}(x, r, \lambda) = \begin{cases} b^T \lambda - \frac{1}{2} \|\lambda\|^2 & \|A^T \lambda\|_\infty \leq \mu \\ -\infty & \text{else} \end{cases}$$

$$(D) \Rightarrow \max_{\lambda} b^T \lambda - \frac{1}{2} \|\lambda\|^2 \quad \text{s.t. } \|A^T \lambda\|_\infty \leq \mu$$

隐式约束: $\min_x c^T x \text{ s.t. } Ax=b, \|x\|_\infty \leq 1$

$$\begin{aligned} \mathcal{L}(x, v) &= c^T x + v^T (Ax - b) \quad . \quad g(v) = \inf_{\|x\|_\infty \leq 1} c^T x + v^T (Ax - b) \\ &= -b^T v - \|A^T v + c\|_1 \end{aligned}$$

适当锥与广义不等式:

Def: 称 K 为适当锥, 若: (i) K 凸 (ii) K 闭 (iii) K 实心 i.e. $\text{int } K \neq \emptyset$ (iv) K 尖: 即 $x \in K \Rightarrow -x \notin K$

Def: $x \preceq_K y \Leftrightarrow y - x \in K$, $x \prec_K y \Leftrightarrow y - x \in \text{int } K$.

Exp: $K = \mathbb{R}_+^n$ 时 即 $\vec{x} \leq \vec{y}$; $K = S_+^n$ 时 $X \preceq_K Y \Leftrightarrow Y - X \succeq 0$

Def: 对偶锥: $K^* = \{y \in \Omega \mid \langle x, y \rangle \geq 0 \quad \forall x \in K\}$

Exp: $(\mathbb{R}_+^n)^* = \mathbb{R}_+^n$, $(S_+^n)^* = S_+^n$ 称 $K^* = K$ 的为自对偶锥

广义不等式约束下 Lagrange 问题:

$$\min_x f(x) \quad \text{s.t.} \quad c_i(x) \preceq_{K_i} 0 \quad i \in I, \quad c_i(x) = 0 \quad i \in E$$

$$\mathcal{L}(x, \lambda, v) = f(x) + \sum_{i \in I} \langle c_i(x), \lambda_i \rangle + \sum_{i \in E} v_i c_i(x) \quad \lambda_i \in K_i^*, \quad v_i \in \mathbb{R}$$

(保证: $\mathcal{L}(x, \lambda, v) \leq f(x)$)

$$g(\lambda, v) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, v), \quad (D): \max_{\lambda_i \in K_i^*, v_i \in \mathbb{R}} g(\lambda, v)$$

半定规划的对偶问题:

$$\min_{X \in S^n} \langle C, X \rangle, \quad \text{s.t.} \quad \langle A_i, X \rangle = b_i, \quad X \succeq 0$$

$$\mathcal{L}(X, y, S) = \langle C, X \rangle - \sum_{i=1}^m y_i (\langle A_i, X \rangle - b_i) - \langle S, X \rangle \quad (S \succeq 0)$$

$$g(y, S) = \begin{cases} b^T y & \sum_{i=1}^m y_i A_i - C + S = 0 \\ -\infty & \text{else} \end{cases}$$

$$(D): \min_y -b^T y \quad \text{s.t.} \quad \sum_{i=1}^m y_i A_i - C + S = 0$$

对偶²: $\min -b^T y \quad \text{s.t.} \quad \sum_{i=1}^m y_i A_i \preceq C$

$$L(y, X) = -b^T y + \langle X, \sum_{i=1}^m y_i A_i - C \rangle, \quad g(X) = \inf_y L(y, X) = \begin{cases} -\langle C, X \rangle & \langle A_i, X \rangle = b_i \quad \forall i \\ -\infty & \text{else} \end{cases}$$

$\Rightarrow (D^2): \min_{X \in S^n} \langle C, X \rangle \quad \text{s.t.} \quad \langle A_i, X \rangle = b_i, \quad X \succeq 0$

Strong Duality: (Slater condition)

- If $p^* > -\infty$, (P) is strictly feasible then (D) is feasible and $p^* = d^*$
- If $d^* < +\infty$, (D) is strictly feasible then (P) is feasible and $p^* = d^*$
- If (P) & (D) has strictly feasible sol., then they both have optimal sol.

Pf: 凸集分离定理 $\Rightarrow$ 构造 $C = \{ (u, v, t) : \exists x, f_i(x) \leq u_i, Ax - b = v, f_0(x) \leq t \}$
 $B = \{ (0, 0, s) : s < p_0^* \}$

$\exists \lambda^T u + \nu^T v + \mu t \geq \alpha \geq \mu p_0^*$, 由 C 定义知 $\lambda \geq 0, \mu \geq 0$, Feasible 保证 $\mu \neq 0$!

